LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 2 Test: Acids, Bases and Salts

Chapter / Topic: Acids, Bases and Salts

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SECTION A

Q1. A solution turns red litmus blue. Its pH is most likely to be: **[1]**

Answer: (d) 10. A solution that turns red litmus blue is basic, so its pH is more than 7.

Q2. Which gas is given off when dilute hydrochloric acid is added to sodium carbonate? **[1]**

Answer: (b) Carbon dioxide. $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$

Q3. The chemical formula of bleaching powder is: **[1]**

Answer: (c) $CaOCl_2$

Q4. Which of the following is used as an antacid? **[1]**

Answer: (b) Milk of magnesia, $Mg(OH)_2$, a mild base that neutralises excess stomach acid.

Q5. Plaster of Paris is obtained by heating: **[1]**

Answer: (d) gypsum ($CaSO_4 \cdot 2H_2O$), heated at 373 K, which gives $CaSO_4 \cdot \frac{1}{2}H_2O$.

Q6. **Assertion (A):** A concentrated acid should be diluted by adding the acid slowly to water, and not water to the acid. **Reason (R):** Mixing a concentrated… **[1]**

Answer: (a). Adding water to the acid can make the mixture splash out and the container may break due to the heat.

SECTION B

Q7. Why does dry HCl gas not change the colour of dry blue litmus paper? **[2]**

Answer: Acids show acidic behaviour only in the presence of water, because H^+ ions (H_3O^+) are produced only in solution. Dry HCl gas does not dissociate into H^+ ions, so it does not affect dry litmus.

- Acids need water to produce H^+ ions: 1
- Dry HCl does not form H^+ ions: 1

OR, answer: A strong acid ionises completely in water and gives a large number of H^+ ions (e.g. HCl, H_2SO_4, HNO_3). A weak acid ionises only partially and gives fewer H^+ ions (e.g. acetic acid, carbonic acid).

- Difference: 1
- One example each: ½ + ½

Q8. Write the chemical name and formula of baking soda. Write the equation for the reaction that takes place when it is heated. **[2]**

Answer: Sodium hydrogencarbonate, $NaHCO_3$. On heating: $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$

- Name and formula: 1
- Balanced equation: 1

SECTION C

Q9. A student tests three solutions P, Q and R with universal indicator and finds their pH to be 2, 7 and 12 respectively. **[3]**

(i) Q (pH 7) [1]

(ii) P. The lower the pH, the higher the H^+ ion concentration. [1]

(iii) R (pH 12), because it is basic. [1]

Q10. What is the chlor-alkali process? Write the balanced equation and give one use of each product. **[3]**

Answer: When electricity is passed through brine (an aqueous solution of sodium chloride), it decomposes to form sodium hydroxide, chlorine and hydrogen: $2NaCl(aq) + 2H_2O(l) \rightarrow 2NaOH(aq) + Cl_2(g) + H_2(g)$. Chlorine is given off at the anode and hydrogen at the cathode. Uses: NaOH for soaps, detergents and paper; Cl_2 for water treatment, PVC and bleaching powder; H_2 for fuels, margarine and ammonia for fertilisers.

- Explanation (electrolysis of brine): 1
- Balanced equation: 1
- One use of each product: 1

SECTION D

Q11. Answer the following: **[5]**

(i) The reaction between an acid and a base to give a salt and water. $NaOH + HCl \rightarrow NaCl + H_2O$ [2]

- Definition: 1
- Balanced equation: 1

(ii) By recrystallisation of sodium carbonate: $Na_2CO_3 + 10H_2O \rightarrow Na_2CO_3 \cdot 10H_2O$. Uses (any one): in the glass, soap and paper industries; for cleaning; to remove permanent hardness of water; to make borax. [2]

- Method/equation with formula: 1
- One use: 1

(iii) They contain acids, which react with the metal to form harmful (poisonous) compounds and also corrode the vessel. [1]

OR, answer: (a) The acid reacts with the metal to form a salt and hydrogen gas, e.g. $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$. Test: bring a burning candle near the gas. It burns with a pop sound, which shows hydrogen. (b) During indigestion the stomach produces too much hydrochloric acid. Antacids are mild bases that neutralise the excess acid. Example: milk of magnesia (magnesium hydroxide).

- (a) statement 1, equation 1, pop test 1
- (b) neutralises excess acid 1, example 1

SECTION E

Q12. Tooth enamel, made of calcium hydroxyapatite (a form of calcium phosphate), is the hardest substance in the body. It does not dissolve in water, but… **[4]**

(i) 5.5 [1]

(ii) To neutralise the excess acid in the mouth and prevent tooth decay. [1]

(iii) Bacteria break down sugar and food particles left in the mouth and produce acids. Prevention (any two): brush after eating, rinse the mouth after meals, avoid sticky sweets, cut down on sugary snacks. [2]

- Source of acid: 1
- Two ways: ½ + ½